

Fantasy Premier League

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The Beautiful Game

This is my journey of playing FPL with 0 ball knowledge and a bit of Python knowledge.

Preseason

The Goal

The purpose of this project is to see how well I can perform using a systematic set of trading rules based on historical data, rather than any knowledge or experience of football

The Goalposts

What does “performing well” mean? The most natural measure of performance is percentile rank among all FPL players, but that doesn’t have an objective benchmark. Is 50% good? 33%?

Perhaps the most objective measure of my performance is the difference between my final point score and the score I would have ended on if I had made no transfers throughout the season. This will evaluate the quality of my transfer decisions, but not of my initial choice of squad. I think I am happy with this, because I expect my initial squad to be far from both the best and worst initial squad.

Since my evaluation method is essentially controlling for the initial squad choice, once the season is over I might be able to work out how well my initial squad would have done with 0 transfers, compared to all the other possibilities.

Relegation zone

That leaves me with a baseline target of ensuring my final points is higher than the points my initial squad would have ended up on. With 37 transfers

Promotion zone

A target that feels certainly ambitious and objective (although, impossible to gauge if at all realistic) target might be to end on more than 1.5x the points that my initial squad would have

Safe zone

Ending up in the Relegation Zone will be an embarrassing but fun story to tell. Ending up in the Promotion Zone will be an even more fun story to tell. If both elude me, then at least I will have the code for my CV.

Formation

How should I choose when to buy, sell, and hold players? And on the basis of what data? I will be using moving averages to make my decisions. I will aim to buy players who are perform well over the last 3 weeks, and vice versa for selling. I will not waste my transfers transacting on averagely performing players, meaning I might not buy good players who are not good enough, and might not sell (i.e. will hold) players who are not (yet) performing badly enough.

Financial fairplay

I will only be using data available from the official FPL API. More specifically, I'll only be using data pertaining to players' points scored in the current and past two seasons. I will not make any decisions based on qualitative or other quantitative data I receive in any form or via any means, external to the API. Secondly, this means I am not "cheating". Primarily, this is because I have no reason to believe I'll be able to employ this information better than that from the API.

Pitch limits

Currently undecided if I want to use chips. I think if I do, I will resign myself to using them the week before they expire, or not at all, on the basis that if I am going to use them, I should use them when I have the most information to hand. But, I don't think I'll use them, because it'll make it very difficult to compare my performance with my initial squad's. I would have to work out how I would have used them, had I kept my initial squad, which feels like a lot of work.

GW 1 - The Initial Squad

With 0 ball knowledge, all I can go off when picking my initial squad is how well they performed last season, and perhaps the season before.

Sample FPL API Data

First, a look at the data we have available from the FPL API.

Players

	PID	first_name	second_name	now_cost
406	337	Brenden	Aaronson	54
571	477	Zach	Abbott	40
52	56	Tammy	Abraham	55
171	151	Josh	Acheampong	45
80	73	Tyler	Adams	50
590	147	Tosin	Adarabioyo	44
647	546	Simon	Adingra	49
84	77	Amine	Adli	49
315	650	David	Affengruber	45
259	633	Honest	Ahanor	45

Teams

	name	played	points	position
0	Arsenal	0	0	1
1	Aston Villa	0	0	19
2	Bournemouth	0	0	15
3	Brentford	0	0	7
4	Brighton	0	0	5
5	Chelsea	0	0	6

	name	played	points	position
6	Coventry City	0	0	20
7	Crystal Palace	0	0	16
8	Everton	0	0	9
9	Fulham	0	0	18

Positions

	singular_name	squad_max_play
0	Goalkeeper	1
1	Defender	5
2	Midfielder	5
3	Forward	3

Combined data

We'll need to combine this data to make use of it, and also declutter what I won't be using. To keep things simple, I'll only be using historical points scored to decide how to change my squad.

	PID	first_name	second_name	team_name	position_name	now_cost	total_points
399	330	Jayden	Bogle	Leeds	Defender	45	37
470	391	Joško	Gvardiol	Man City	Defender	57	33
181	165	João Pedro	Junqueira de Jesus	Chelsea	Forward	78	33
133	124	Pascal	Groß	Brighton	Midfielder	56	33
487	411	Erling	Haaland	Man City	Forward	155	33
124	115	Maxim	De Cuyper	Brighton	Defender	48	32
11	12	Bukayo	Saka	Arsenal	Midfielder	95	30
104	94	Kevin	Schade	Brentford	Midfielder	60	30
268	229	James	Tarkowski	Everton	Defender	61	29
75	68	Marcus	Tavernier	Bournemouth	Midfielder	61	29
513	426	Bruno	Borges Fernandes	Man Utd	Midfielder	120	29
0	1	David	Raya Martín	Arsenal	Goalkeeper	60	29
443	367	Cody	Gakpo	Liverpool	Midfielder	72	29
347	572	Konstantinos	Tzolakis	Hull City	Goalkeeper	46	28
7	8	Riccardo	Calafiori	Arsenal	Defender	58	28
76	69	Alex	Scott	Bournemouth	Midfielder	61	28
14	15	Martin	Ødegaard	Arsenal	Midfielder	67	27

	PID	first_name	second_name	team_name	position_name	now_cost	total_points
330	286	Mohamed	Belloumi	Hull City	Midfielder	50	27
158	40	Morgan	Rogers	Chelsea	Midfielder	77	27
173	154	Cole	Palmer	Chelsea	Midfielder	97	26

Prior Season Data

The ‘total_points’ column will update each week. What I need for my initial squad selection are players’ total points from the last 2 seasons.

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost
0	1	David	Raya Martín	Goalkeeper	60	304	5.066667
1	2	Kepa	Arrizabalaga Revuelta	Goalkeeper	49	108	2.204082
2	3	Illan	Meslier	Goalkeeper	49	0	0.000000
3	4	Gabriel	dos Santos Magalhães	Defender	80	326	4.075000
4	5	Jurriën	Timber	Defender	65	263	4.046154
5	6	William	Saliba	Defender	59	267	4.525424
6	7	Myles	Lewis-Skelly	Midfielder	55	66	1.200000
7	8	Riccardo	Calafiori	Defender	58	148	2.551724
8	9	Piero	Hincapié	Defender	53	87	1.641509
9	10	Benjamin	White	Defender	55	95	1.727273
10	11	Cristhian	Mosquera	Defender	54	40	0.740741
11	12	Bukayo	Saka	Midfielder	95	284	2.989474
12	13	Declan	Rice	Midfielder	74	311	4.202703
13	14	Eberechi	Eze	Midfielder	63	261	4.142857
14	15	Martin	Ødegaard	Midfielder	67	190	2.835821
15	16	Noni	Madueke	Midfielder	62	192	3.096774
16	17	Mikel	Merino Zazón	Midfielder	58	159	2.741379
17	18	Gabriel	Martinelli Silva	Midfielder	63	182	2.888889
18	19	Martín	Zubimendi Ibáñez	Midfielder	53	133	2.509434
19	20	Max	Dowman	Midfielder	53	16	0.301887

The Initial Squad

Before I choose which players I want, how do I choose how many of each position I want, subject to the maximum constraints? I’ll decide based on the average points scored for each position, over the last season:

Mean Goalkeeper Points over last season: 72
 Mean Defender Points over last season: 74
 Mean Midfielder Points over last season: 70
 Mean Forward Points over last season: 70
 Mean Goalkeeper cost: 4.8
 Mean Defender cost: 4.9
 Mean Midfielder cost: 5.6
 Mean Forward cost: 6.1

So, very crudely, Defenders are a bit more lucrative than other positions, and the opposite for Forwards. So I'll pick two Goalkeepers, then as many Defenders as I can, then as many Midfielders as I can, and then as few Forwards as I can. Perhaps this formation will change as the season progresses, depending on whether relative points/cost changes between different positions. I also will use as a tiebreaker a preference to add players from teams not already in my squad, to hopefully reduce the variance of my weekly points.

Below, we'll order players by the ratio of their points over the last two seasons, divided by their current cost. I'll then select for my initial squad players with a high ratio and a low cost. The idea is that I'm happy to buy underpriced players who score fewer points, so that I have the flexibility later in the season to use current season data to make transfers. Currently, I'm picking based on prior season performance, which it feels fair to assume is uniformly a worse indicator for future performance than current season data.

As we only have current cost, which is a dynamic field, the tables below are likely to show different players than the ones I chose at the time of writing. But, the logic is the same.

Goalkeepers

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
265	226	Jordan	Pickford	Goalkeeper	55	293	5.327273
229	198	Dean	Henderson	Goalkeeper	50	266	5.320000
288	250	Bernd	Leno	Goalkeeper	45	237	5.266667
120	109	Bart	Verbruggen	Goalkeeper	45	233	5.177778
561	467	Matz	Sels	Goalkeeper	50	255	5.100000
0	1	David	Raya Martín	Goalkeeper	60	304	5.066667
161	140	Robert	Lynch Sánchez	Goalkeeper	49	239	4.877551
157	28	Emiliano	Martínez Romero	Goalkeeper	50	231	4.620000
530	442	Nick	Pope	Goalkeeper	49	198	4.040816
93	82	Caoimhín	Kelleher	Goalkeeper	50	188	3.760000

I choose Verbruggen and Leno

Defenders

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
233	204	Tyrick	Mitchell	Defender	45	258	5.733333
560	201	Daniel	Muñoz Mejía	Defender	54	278	5.148148
467	388	Marc	Guéhi	Defender	60	295	4.916667
432	356	Virgil	van Dijk	Defender	65	318	4.892308
272	233	Vitalii	Mykolenko	Defender	45	216	4.800000
565	471	Nikola	Milenković	Defender	55	264	4.800000
589	112	Jan Paul	van Hecke	Defender	49	235	4.795918
563	469	Neco	Williams	Defender	50	237	4.740000
95	84	Nathan	Collins	Defender	55	256	4.654545
5	6	William	Saliba	Defender	59	267	4.525424
26	31	Ezri	Konsa Ngoyo	Defender	45	203	4.511111
268	229	James	Tarkowski	Defender	61	275	4.508197
185	200	Maxence	Lacroix	Defender	60	268	4.466667
567	473	Ola	Aina	Defender	45	195	4.333333
31	32	Matty	Cash	Defender	45	193	4.288889
536	448	Dan	Burn	Defender	49	208	4.244898
504	417	Diogo	Dalot Teixeira	Defender	49	206	4.204082
164	143	Trevoh	Chalobah	Defender	55	229	4.163636
30	30	Lucas	Digne	Defender	45	186	4.133333
60	611	Aaron	Wan-Bissaka	Defender	45	185	4.111111

I choose Mitchell, Guehi, Van Hecke, Tarkowski, and N.Williams

Midfielders

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_per_cost_2yr
514	427	Bryan	Mbeumo	Midfielder	79	384	4.860759
297	261	Alex	Iwobi	Midfielder	54	259	4.796296
104	94	Kevin	Schade	Midfielder	60	274	4.566667
488	481	Elliot	Anderson	Midfielder	63	287	4.555556
106	96	Mikkel	Damsgaard	Midfielder	55	244	4.436364
476	397	Antoine	Semenyo	Midfielder	84	367	4.369048
574	480	Morgan	Gibbs-White	Midfielder	79	342	4.329114
444	368	Dominik	Szoboszlai	Midfielder	70	303	4.328571
27	452	Bruno	Guimarães Rodriguez Moura	Midfielder	67	289	4.313433
237	208	Ismaila	Sarr	Midfielder	62	266	4.290323

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_pe
158	40	Morgan	Rogers	Midfielder	77	330	4.285714
448	372	Alexis	Mac Allister	Midfielder	54	230	4.259259
462	155	Enzo	Fernández	Midfielder	69	292	4.231884
12	13	Declan	Rice	Midfielder	74	311	4.202703
13	14	Eberechi	Eze	Midfielder	63	261	4.142857
463	237	Iliman	Ndiaye	Midfielder	59	242	4.101695
543	457	Jacob	Murphy	Midfielder	59	241	4.084746
515	428	Matheus	Santos Carneiro da Cunha	Midfielder	79	321	4.063291
447	371	Ryan	Gravenberch	Midfielder	59	236	4.000000
623	525	Mateus	Fernandes	Midfielder	58	230	3.965517

I choose Szoboszlai, Rice, Anderson, Iwobi, and Damsgaard

Forwards

	PID	first_name	second_name	position_name	now_cost	points_2yr	points_
51	55	Ollie	Watkins	Forward	78	353	4.52564
160	136	Danny	Welbeck	Forward	59	251	4.25423
251	223	Jean-Philippe	Mateta	Forward	63	264	4.19047
583	490	Chris	Wood	Forward	58	241	4.15517
86	79	Francisco Evanilson	de Lima Barbosa	Forward	60	243	4.05000
181	165	João Pedro	Junqueira de Jesus	Forward	78	303	3.88461
250	222	Jørgen	Strand Larsen	Forward	60	220	3.66666
284	248	Norberto Bercique	Gomes Betuncal	Forward	54	191	3.53703
550	464	Yoane	Wissa	Forward	62	212	3.41935
414	346	Dominic	Calvert-Lewin	Forward	60	205	3.41666
134	125	Georginio	Rutter	Forward	54	182	3.37037
559	167	Liam	Delap	Forward	55	177	3.21818
455	379	Alexander	Isak	Forward	91	252	2.76923
487	411	Erling	Haaland	Forward	155	420	2.70967
625	527	Richarlison	de Andrade	Forward	59	154	2.61016
624	526	Dominic	Solanke-Mitchell	Forward	58	148	2.55172
114	106	Igor Thiago	Nascimento Rodrigues	Forward	79	190	2.40506
306	271	Rodrigo	Muniz Carvalho	Forward	54	122	2.25925
584	491	Igor Jesus	Maciel da Cruz	Forward	58	114	1.96551
527	440	Joshua	Zirkzee	Forward	54	106	1.96296

I choose Calvert-Lewin, Mateta, and Welbeck

Leftover Capital

This leaves me with £14m to play with. If I make 38 transfers over the season, I can average a net cost of ~£0.5m per transfer. The idea is that every transfer I make will use some capital to improve my team made from old data, on the basis of new data (current season data). I expect my initial squad to optimal with respect to the old data (as far as my selection logic is concerned), but to be a very poor predictor of this season's performance when compared with current season data, as it eventually emerges. Selecting a cheap initial squad gives me flexibility to leverage current season data to make "corrections" to my initial squad that was chosen from old data.